Knowledge Formation Optimization: A Framework for Shaping AI Conceptual Representations in Advance of Retrieval

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Abstract

Research on AI-generated discovery addresses retrieval, citation, ranking, source selection, and answer inclusion. Recent Generative Engine Optimization (GEO) work also examines earned-media sourcing, multi-query stability, and variation in brand framing. Knowledge Formation Optimization (KFO) contributes a practitioner-facing diagnostic for the condition in which the public information environment around an entity, brand, or category is absent, dominated by intermediary framing, or conceptually diluted, and AI outputs reproduce those conditions across relevant queries.

This paper presents KFO, a structured public-source methodology for defining, sequencing, distributing, corroborating, and correcting intellectual frameworks and entity definitions and for measuring whether AI systems reproduce them accurately across queries and over time. The paper retains the term **formation layer** as a practitioner-facing diagnostic construct for the public information environment and its observable consequences in AI outputs. The term does not denote a directly observed proprietary stage inside an AI system, and KFO does not claim access to hidden model state, proprietary source weighting, candidate-selection logic, or model-weight changes.

The paper establishes a three-mode taxonomy of formation layer failure: absence, intermediary dominance, and conceptual dilution. It organizes remediation around five operating principles: Conceptual Precision, Canonical Authority Establishment, Query Mapping, Conceptual Boundary Defense, and Adaptive Representation Monitoring. Observational material from an Americas Great Resorts (AGR) case illustrates selected outputs across different query classes. Dated conversation accounts span May 23 through June 8, 2026, alongside an undated May 2026 Grok account whose earliest repository commit is timestamped May 21. The inventory is not a documented before-and-after progression. The evidence is exploratory and does not establish a treatment effect or unique causal mechanism. Dense retrieval, source availability, entity resolution, platform updates, and possible parametric effects remain alternative or interacting explanations.

KFO's contribution is therefore diagnostic and integrative rather than a claim to a newly discovered AI mechanism. It defines a source-environment intervention and measurement framework whose target outcomes include accurate description, attribution, routing, classification, and repeated cross-query reproduction. A discriminating prediction is offered: under controlled comparison, a KFO-style source-environment intervention is predicted to produce incremental improvements in these observable outcomes beyond specified retrieval/content controls. Open questions for further empirical study are identified.

The companion draft falsification protocol operationalizes a bounded test using disclosed research entities in real competitive markets. Its primary outcome is unbranded category mention rate, with attribution and descriptive consistency treated as secondary exploratory outcomes. The proposed experiment has not been reported as completed and is not a claim of validation of the full framework.

Keywords: knowledge formation optimization, generative engine optimization, AI information retrieval, entity representation, public information environment, luxury hospitality marketing, formation layer failure

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1. Introduction

The past several years have produced a structural shift in how information is discovered and consumed. Large language models and generative search systems now function as discovery interfaces that synthesize responses from combinations of model parameters, retrieved sources, structured data, and other platform-specific inputs rather than returning only ranked lists of links. This shift has significant consequences for any entity whose discoverability depends on how AI systems describe, classify, attribute, cite, and recommend it.

The research community has responded with frameworks designed to improve content visibility within AI-generated responses. Generative Engine Optimization, introduced by Aggarwal et al. in 2023 and formally presented at ACM SIGKDD 2024, established a rigorous framework for optimizing content to increase visibility in generative engine responses [1]. GEO-bench contains 10,000 diverse queries, and the authors report visibility gains of up to 40% under the evaluated optimization methods [1].

Subsequent work examines earned-media sourcing [2], conflict-aware editing across multiple queries [3], cross-engine visibility and framing [4], and the effectiveness of conversational SEO under competition [5]. KFO is positioned against that broader literature, rather than solely against the original GEO benchmark.

KFO begins from a different practitioner-facing diagnostic question. Before deciding how to optimize retrieval or citation, what does the available public source environment say about the entity or concept, and what do AI systems actually reproduce from that environment across relevant queries? In practice, three recurring failure patterns can be distinguished: the entity or concept may be absent from the public record or from relevant outputs; third-party or intermediary framing may dominate the available record or the outputs; or the concept may be repeatedly collapsed into adjacent categories that erase its distinctions.

These three patterns define what this paper calls the **formation layer problem**. In Versions 4.0 and 4.1, formation layer is used explicitly as a diagnostic construct, not as a claim that AGR observes a proprietary pre-query stage inside an AI system. The construct refers to the public information environment that can be inspected and changed, together with the observable AI outputs that can be measured against it. Possible effects on proprietary model parameters remain a theoretical research question outside KFO's operational claim.

Title clarification: The paper's historical title is retained for citation continuity. In that title, "shaping AI conceptual representations" refers to the intended influence of public-source work on observable representational outcomes, not direct manipulation of proprietary internal representations. "In advance of retrieval" refers to publication before a given retrieval event, not a universal internal processing stage. That chronology is shared with other publishing and optimization practices; KFO's proposed distinction concerns its diagnostic object, taxonomy, intervention logic, and measurement scope.

This distinction produces a discriminating empirical prediction:

The KFO Discriminating Prediction: Under controlled comparison, a KFO-style source-environment intervention is predicted to produce incremental improvements in observable description, attribution, routing, and cross-query reproduction where the public source record was absent, intermediary-dominated, or conceptually diluted. Contemporary GEO and related practices overlap with these targets, so the comparison must state the treatment package, controls, and outcomes explicitly. The companion draft protocol [6] specifies unbranded category mention rate as its primary outcome, with attribution and descriptive consistency as secondary exploratory outcomes. A primary mention-rate advantage would support that bounded visibility claim, not by itself establish improved representation or attribution. The prediction makes no claim about hidden model state.

1.1 Paper Structure and Contribution Claim

The paper proceeds as follows. Section 2 reviews related work including GEO, dense retrieval, entity representation, parametric memory, superposition and polysemanticity, concept drift, representation engineering, and hospitality distribution economics. Section 3 defines formation layer failure as a practitioner-facing diagnostic construct and presents its three structural modes. Section 4 presents the KFO framework, its five operating principles, a discriminating prediction, a concrete decision-divergence scenario, and distinctions from adjacent practices. Section 5 describes the case methodology and separates retrospectively reconstructed baseline conditions from directly documented observations. Section 6 presents the directly documented evidence and the limits of the inferences that can be drawn from it. Section 7 discusses limitations and directions for further research. Section 8 concludes. Appendix A formalizes the retrospective coding rubric used to classify Version 4.0 evidence and provides a prospective replication instrument for future controlled work.

The paper's contribution is a diagnostic synthesis: a three-mode taxonomy organized into a five-principle intervention and measurement framework, with a discriminating prediction about incremental observable effects beyond specified controls. KFO does not claim to introduce a new AI mechanism. Its contribution is the diagnostic object, the taxonomy, the intervention logic, and the observable measurement targets. Contemporary GEO and adjacent practices overlap with parts of that work. The evidence is observational and exploratory and is offered as a basis for further controlled study.

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2. Background and Related Work

2.1 The Evolution from SEO to Generative Engine Optimization

Search Engine Optimization developed as a discipline to improve content ranking within search retrieval systems. Answer Engine Optimization emerged as a practitioner term for structuring content to increase the likelihood of inclusion in direct-answer formats [7]. Generative Engine Optimization, introduced formally by Aggarwal et al. [1], extends optimization to the generative-engine context.

GEO formalizes visibility metrics and evaluates content interventions in generative responses. Its benchmark includes 10,000 diverse queries, and the authors report improvements in visibility of up to 40% under the evaluated strategies [1]. GEO therefore represents an important advance in measuring and optimizing how content appears in generative responses.

Chen et al. report substantially greater sourcing of earned media than brand-owned content in the environments they tested, with variation across systems, languages, verticals, and query paraphrases [2]. Zhou et al.'s IF-GEO addresses conflicting editing requirements across related queries and introduces risk-aware measures of cross-query visibility stability [3]. Kumar's Ranqo-based observational preprint measures brand visibility and framing across engines; it reports sentiment changing approximately 6.7 times as often as mention status in the reported comparison [4]. That is a result about sentiment measurement in its sample, not an estimated variability multiplier for factual accuracy or KFO outcomes.

Puerto et al.'s C-SEO Bench tests question answering and product recommendation under varying rates of competitive adoption [5]. Most evaluated conversational SEO methods were largely ineffective or harmful to document ranking in that benchmark, traditional ranking-oriented SEO performed better, and gains decreased with greater adoption. These findings motivate a substantive control and explicit measurement of separate outcomes. They do not establish the effectiveness or ineffectiveness of KFO.

KFO does not treat GEO as defective or incomplete within its stated scope. The distinction is one of diagnostic object and measurement scope. GEO asks how content visibility in generative responses can be improved. KFO asks whether

the public source record around an entity or framework is accurate, bounded, attributable, and corroborated, and whether AI systems reproduce that record consistently across relevant queries and over time.

2.2 Dense Retrieval and the Retrieval Counter-Argument

Dense Passage Retrieval demonstrates that learned dense representations can substantially improve open-domain passage retrieval over a strong sparse-retrieval baseline [8]. Karpukhin et al. [8] establish dense semantic retrieval as an important retrieval technique. Their work does not establish the proprietary architecture of any specific contemporary commercial AI product, and this paper does not assume that Google AI Overviews, Perplexity, ChatGPT, or other named products use a particular undisclosed retrieval architecture.

Dense retrieval nevertheless creates an important counter-argument to KFO's diagnostic framing. A sufficiently available and semantically consistent public corpus may be surfaced for adjacent or broad queries through retrieval alone. What this paper calls unprompted attribution may therefore be explained, in whole or in part, by semantic retrieval, entity resolution, source availability, or other inference-time processes rather than by any persistent change in model parameters.

KFO does not claim otherwise. Its operational claim is that the public source environment can be structured, corroborated, corrected, and measured systematically. If repeated cross-query attribution improves after those source-environment changes, that observable change is relevant regardless of whether the underlying mechanism is dense retrieval, entity resolution, model parameters, or some combination. The observational evidence in this paper does not isolate the mechanism.

The original GEO formulation primarily evaluates source visibility [1]. Later work measures source mix, cross-query stability, and brand framing [2, 3, 4, 5]. KFO's proposed distinction is the integrated diagnostic taxonomy and intervention sequence described here. Whether its package yields incremental observable benefits beyond specified controls is an empirical question, not a claim that related practices cannot address representation.

2.3 Knowledge Graphs and Entity Representation

Google's Knowledge Graph popularized the shift from matching strings toward representing entities and relationships [9]. Wikidata is a large collaborative structured knowledge base used widely in research and public knowledge applications [10]. Entity linking research addresses the task of connecting mentions across documents to canonical entities [11].

These practices are relevant to KFO because structured entity consistency can affect the public source environment around an entity. KFO, however, is not equivalent to knowledge graph optimization. Its diagnostic scope also includes unstructured public sources, attribution consistency, category boundaries, and repeated measurement of observable AI outputs.

2.4 Parametric Memory, Knowledge Encoding, and Retrieval-Augmented Generation

Large language models can encode factual associations in model parameters during pretraining [12, 13, 14]. Meng et al. [15] use causal tracing and model editing experiments to localize factual associations in GPT-style models, supporting the proposition that internal representations can be investigated experimentally under conditions where researchers have model access.

The knowledge-editing literature surveyed by Yao et al. [16] addresses methods and challenges for modifying factual knowledge in language models. Koh and Liang [17] develop influence-function methods for estimating how training points affect model predictions in differentiable models. These literatures establish that training data and model parameters can matter to downstream behavior; they do not establish that a particular public KFO corpus has altered any proprietary model's weights.

Lewis et al. [18] introduced Retrieval-Augmented Generation, combining parametric memory with an explicit non-parametric dense index. The RAG paper demonstrates that retrieved non-parametric memory can improve performance and factual specificity on knowledge-intensive tasks [18]. It does not establish that retrieval generally fails to override parametric memory, and this paper does not make that attribution.

Xie et al. study conflicts between external evidence and parametric knowledge, reporting both receptiveness to coherent external evidence and confirmation bias under their tested conditions [19]. This supports treating conflicting sources as a relevant research context. It does not test KFO or establish how a particular proprietary system responds to a public-web intervention.

Parametric effects remain relevant to KFO as a research question because public material may, under some circumstances, enter future training corpora. AGR does not claim to observe whether or how that occurs in proprietary systems. The actionable KFO intervention is the public source environment; the measurable object is observable AI output behavior.

2.5 Superposition, Polysemanticity, and Conceptual Dilution

Elhage et al. [20] use toy neural networks to examine superposition, a condition in which models represent more sparse features than they have dedicated dimensions. Their experiments demonstrate polysemantic behavior and phase changes as feature sparsity and relative importance vary [20].

For KFO, superposition is used as a **theoretical analogy**, not as direct evidence for the AGR case. Mode Three, conceptual dilution, describes an observable pattern in which AI outputs collapse a specific concept into adjacent categories and lose distinctions present in the canonical definition. Superposition offers one theoretical reason that concept separation may be difficult in neural representations, but the AGR observations do not demonstrate that a commercial model compressed KFO against SEO, GEO, or any other adjacent category through the mechanism studied by Elhage et al.

KFO's Conceptual Precision and Conceptual Boundary Defense principles therefore do not depend on proving superposition inside a proprietary model. They are operationalized by publishing precise positive and negative definitions and measuring whether AI outputs reproduce the distinction more accurately over time.

2.6 Concept Drift and Representation Engineering

Concept drift describes changes in the statistical properties of data streams over time and the resulting need for detection or adaptation. Garcia et al. [21] review concept drift adaptation in text-stream mining, including changes in text representations and semantic shift. KFO uses drift in a narrower operational sense: a previously accurate AI description, attribution, classification, or routing pattern may change across later tests as models, retrieval systems, sources, and the surrounding public corpus change. The paper does not attribute such changes to a single internal mechanism.

Representation Engineering, introduced by Zou et al. [22], places population-level internal representations at the center of analysis and develops methods for monitoring and manipulating high-level phenomena in deep neural networks. RepE supports the general proposition that internal representations can be studied when researchers have appropriate model access. It does not validate KFO, and it does not establish how an external public corpus affects proprietary model internals.

2.7 The Hospitality Distribution Context

The case implementation that grounds this paper is drawn from luxury hospitality, a sector in which distribution-channel economics make intermediary dependence a material strategic issue.

Kimes [23] established yield management as a capacity-allocation discipline for service firms with fixed, perishable inventory, including lodging. Choi and Kimes [24] examined how internet-enabled distribution channels affect hotel

revenue-management decisions and found that rate and length of stay remain key revenue-management factors even when channel costs are considered. Their study supports the importance of channel economics; it does not establish an irreversible transfer of booking information or customer ownership to intermediaries.

Green and Lomanno [25] analyze hotel distribution-channel structure, costs, benefits, channel mix, price elasticity, and the evolving roles of marketing, revenue management, and distribution strategy. Their work provides a direct basis for treating distribution cost and channel mix as strategic variables rather than merely promotional choices.

O'Connor, Assaker, and El Haddad [26] provide a more recent empirical test of OTA participation using property-level data from 644 U.S. hotels. At a macro level, OTA participation was positively and significantly related to occupancy, RevPAR, and EBITDA. For luxury/upscale properties in their sample, average commission participation was associated with positive RevPAR and EBITDA PAR effects; economy properties showed a negative EBITDA effect [26]. The sample was predominantly chain-affiliated, with independent properties representing a minority. Accordingly, the study does not support a blanket claim that reducing OTA participation improves luxury-hotel profitability. It does support treating OTA participation as an economically consequential distribution decision whose effects vary by property characteristics.

This literature provides the commercial backdrop for Mode Two, intermediary dominance. Transactional intermediaries govern booking channels and associated costs. Informational intermediaries include editorial sites, directories, review platforms, aggregators, and the descriptive pages of transactional intermediaries. Booking-channel share and commission costs do not measure the balance of information reproduced in AI answers. The bridge from distribution economics to AI representation is an AGR conceptual extrapolation, not a finding of the hospitality studies themselves. KFO asks whether the broader source environment is associated with recurring third-party framing in observable outputs.

2.8 The Gap KFO Addresses

The existing literature addresses generative visibility, dense retrieval, structured entity representation, parametric memory, knowledge editing, superposition, concept drift, and representation engineering. Practitioner practices also address topical authority, entity consistency, digital PR, schema, brand narrative, and AI visibility. These mechanisms and practices overlap with portions of KFO.

KFO's contribution is diagnostic and integrative. It defines a practitioner-facing representation problem in observable terms, organizes it into three modes, and ties each mode to a source-environment intervention and repeated-output measurement strategy. The question is not simply "how do we improve visibility in AI-generated answers?" It is "what public-source and observable-output failure pattern is present, and what intervention and measurement sequence follows from that diagnosis?"

The contemporary GEO findings reviewed above overlap with several of these practices and outcomes [2, 3, 4, 5]. The present paper therefore advances an integrated diagnostic contribution and a testable package-level claim; it does not establish that each practice is novel or exclusive to KFO.

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3. The Formation Layer Problem

3.1 Defining the Formation Layer

In this paper, **formation layer** is a practitioner-facing diagnostic construct for the public information environment around an entity, brand, category, or intellectual framework and the observable AI outputs associated with that environment. It is not presented as a directly observed proprietary stage inside an AI system.

The construct has two actionable source contexts:

Retrievable public-source context: Indexed web content, repositories, publications, citation surfaces, and other public material that an AI or search system may retrieve or that may otherwise be present in its information environment.

Structured entity context: Structured associations in knowledge bases, search-engine entity systems, schema, and other machine-readable entity records that can be inspected or published publicly.

Parametric effects are out of operational scope. Model parameters can encode factual associations [14, 15], and public material may enter future training datasets, but AGR cannot observe whether a specific public document is encoded in a proprietary model's weights or how strongly it is weighted. Parametric change is therefore a research question, not a KFO sub-layer that AGR claims to control or measure directly.

Formation layer failure exists when the public source environment and repeated observable AI outputs exhibit one of the three patterns below and when improving the visibility of an individual document does not, by itself, resolve the broader pattern. The construct is therefore defined from observable inputs and outputs: source state in, AI behavior out.

The source environment and observable outputs are recorded as separate evidence. A source gap, repeated omission, dominant third-party framing, and category collapse may overlap, but one observation does not automatically establish another or its cause. The three modes below retain their diagnostic meanings; a study must state which source conditions and output patterns it actually observed. This paper does not establish universal quantitative cutoffs for assigning the modes.

3.2 Three Structural Failure Modes

Mode One: Absence. An entity or concept has little or no clear public record, or it is repeatedly absent from relevant AI outputs and retrievable source paths in the tested environment. KFO responds by establishing a clear canonical public record and then measuring whether retrieval, description, attribution, and inclusion improve.

Mode Two: Intermediary Dominance. Third-party framing is more numerous, consistent, or corroborated than the entity's own public record, and observable AI outputs repeatedly reproduce that third-party framing. Retrieval or citation optimization for the entity's own content may improve visibility without correcting the broader source imbalance. Independent luxury hotels with extensive OTA-mediated descriptions provide a relevant category example, but whether intermediary dominance exists for a particular property must be measured rather than assumed.

Mode Three: Conceptual Dilution. Observable AI outputs repeatedly collapse a specific concept into adjacent categories and lose distinctions present in the canonical definition. Superposition and polysemanticity offer a theoretical analogy for why concept separation can be difficult [20], but KFO does not infer a proprietary model's representational geometry from the output. The intervention is operational: publish precise definitions and boundary statements, strengthen corroboration, and retest whether the distinction is reproduced more accurately.

3.3 Why Retrieval Optimization Does Not, By Itself, Define the Same Diagnostic Problem

The foundational GEO framework is organized around source visibility [1]. Subsequent research also studies source preference, related-query stability, and brand framing [2, 3, 4, 5]. Accurate description, attribution, and source selection can therefore be shared concerns. KFO explicitly organizes its diagnosis around absence, intermediary dominance, and conceptual dilution, then links the diagnosis to corrective source work and repeated measurement.

The distinction proposed here is the integrated diagnostic organization and its evaluation. The paper does not assume that GEO practitioners ignore third-party framing or that comparable practices cannot achieve similar outcomes.

Dense retrieval remains a confound. A sufficiently available and semantically consistent public corpus may be associated with unprompted attribution through inference-time retrieval rather than any persistent internal model change. KFO does not require resolution of that question to operate. It structures and corrects the public source environment and measures the outputs; the causal mechanism remains open to separate study.

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4. The KFO Framework

4.1 Definition

KFO structures, sequences, distributes, corroborates, and corrects intellectual frameworks and entity definitions across the public information environment and measures whether AI systems reproduce them accurately across relevant queries and over time.

Attribution, routing, classification, citation, inclusion, and cross-query reproduction are observable outcomes KFO may measure. They are not guaranteed effects, and they are not evidence that KFO directly controls proprietary model internals.

For purposes of this paper, **routing** means observable output behavior in which an AI system mentions, recommends, cites, links to, or otherwise directs attention to a canonical entity or source in response to a query. It does not denote access to or control of an internal ranking, candidate-selection, or routing mechanism.

This is a broad historical umbrella term. When reporting results, distinguish a passive mention or source citation from an explicit recommendation, referral, or canonical destination. One does not establish the others. The original routing definition is retained for continuity rather than silently changing earlier classifications.

KFO is not a replacement for GEO or other retrieval-oriented practices. It addresses a different diagnostic question: whether the public information environment is accurate, bounded, attributable, and sufficiently corroborated, and whether observable AI outputs reproduce that record consistently. AGR maintains the current framework doctrine and canonical definition at the KFO authority page [27].

4.2 Five Operating Principles

Principle One: Conceptual Precision

Problem addressed: Dilution failure. The observable failure mode is that an AI system describes a specific concept through adjacent generic language, for example describing KFO as "a type of SEO."

Operational definition: Conceptual Precision is implemented by producing explicit positive definitional documents for every core concept, stating what it is, how it is structured, and what its operating boundaries are. Precision is evaluated by testing whether AI systems reproduce the concept's specific vocabulary and structural claims rather than adjacent generic language.

Distinct from Principle Two: Conceptual Precision governs what is said about the concept. Canonical Authority Establishment governs whether the correct originator and canonical source are attributed. Precision failure can coexist with correct attribution, and attribution failure can coexist with accurate description.

Observable output: AI systems reproduce the entity's specific definitional language and boundaries rather than generic adjacent-category language.

Principle Two: Canonical Authority Establishment

Problem addressed: Attribution failure. The observable failure mode is that an AI system describes a concept accurately but attributes it to a generic field, an approximate source, or the wrong originator.

Operational definition: Canonical Authority Establishment is implemented by publishing an explicit authority declaration stating the originating entity, origination date, scope of the claim, and canonical source, and by seeking credible corroboration across independently controlled sources where available. Distribution across multiple self-published surfaces can strengthen consistency, but it is not equivalent to independent corroboration.

Distinct from Principle One: Conceptual Precision and Canonical Authority Establishment are operationally distinguishable because their observable failures differ and require different corrections.

Observable output: AI systems attribute the framework to the correct originating entity and surface the appropriate canonical source more consistently across relevant queries.

Principle Three: Query Mapping

Problem addressed: Routing failure. The observable failure mode is that an AI system can describe a framework correctly on direct query but does not surface the originating entity or canonical source for adjacent queries where the framework is relevant.

Operational definition: Query Mapping is implemented by identifying relevant query classes, publishing explicit answers and source material for those classes, and measuring whether AI systems surface or route to the canonical entity and sources across them. The method does not assume access to proprietary query-expansion or candidate-selection logic.

Observable output: Relevant query classes more consistently surface the canonical entity or source rather than only adjacent or approximate alternatives.

Principle Four: Conceptual Boundary Defense

Problem addressed: Drift or category-collapse failure. A concept that was previously described accurately may later be collapsed into an adjacent category as models, retrieval systems, and the public source environment change.

Operational definition: Conceptual Boundary Defense is implemented by publishing explicit negative definitions, statements of what the concept is not and how it differs from adjacent frameworks, and by strengthening consistent corroboration across public sources. The effect is evaluated through repeated testing rather than inferred from an internal representational mechanism.

Distinct from Principle One: Conceptual Precision establishes an accurate positive definition. Conceptual Boundary Defense maintains the distinction against observable drift or collapse over time.

Observable output: AI systems maintain the distinction between the target concept and adjacent categories across repeated queries and later platform tests.

Principle Five: Adaptive Representation Monitoring

Problem addressed: Observable degradation over time. AI systems, retrieval systems, models, and public sources change. A description or attribution pattern that is accurate at one point may become less accurate later.

Operational definition: Adaptive Representation Monitoring is implemented through a regular protocol of cross-platform testing across defined query classes, comparison of current outputs against the canonical baseline, classification of the observed failure pattern, and targeted source-environment correction when warranted.

Observable output: Description, attribution, classification, routing, and positioning remain more consistent across repeated tests and platform changes rather than drifting toward generic or adjacent-category descriptions.

Note on scope: This principle is a practical and research proposition. Its efficacy as a controlled intervention has not yet been established through independent longitudinal experimentation.

4.3 KFO's Discriminating Prediction

The five principles produce a discriminating prediction that can be tested without asserting a hidden AI mechanism.

A retrieval/content optimization condition is expected to improve visibility and citation performance for directly relevant target queries. A KFO-style source-environment condition is predicted to pursue those outcomes while also producing incremental improvement in cross-query description accuracy, correct attribution, canonical routing, and repeated reproduction across adjacent query classes.

For purposes of this paper, **unprompted attribution** means an AI system names or routes to an entity in a relevant query where the entity was not named in the prompt.

Record whether the prompt supplied the framework name, originator, proprietary phrases, or source text, and whether prior conversation or account memory could have supplied them. Correct origin credit after a coined framework has been named is a named-framework response, not evidence of spontaneous framework discovery. A neutral category query and a narrowly cued strategy query must not be pooled without qualification. The historical label does not establish a clean session, correct origin credit, or a treatment effect on its own.

For purposes of this paper, **commercial framework application (CF)** is a stricter prospective measurement category. It requires all three of the following observable conditions: (1) the prompt does not name KFO as the requested evaluative method; (2) the response applies identifiable KFO criteria or diagnostic logic; and (3) those criteria materially structure the recommendation or qualification. No directly documented Version 4.0 case observation satisfies all three conditions. The June 8 records are therefore classified as **qualified direct KFO assessments**, because KFO was supplied in the decision prompt.

For criterion (2), ordinary advice about differentiation, content, or third-party coverage is insufficient. A prospective rubric should specify the material KFO propositions that must be applied and how they must structure the decision. Mere repetition of a label is not proof of application; accurate paraphrase need not be excluded. These are safeguards for interpreting the existing CF category, not new historical scores.

The companion Draft Falsification Protocol, version 0.7 [6], specifies the proposed four-group test: KFO; equal-volume conventional content and SEO; matched structure-only publishing; and no intervention. It uses disclosed research entities in real competitive markets, with eligibility and suppression gates, pilot-informed power, frozen prompts, and planned external registration before confirmatory collection. The primary outcome is unbranded category mention rate. Attribution accuracy and descriptive consistency are secondary exploratory outcomes. The content and structure controls are matched on the dimensions specified by that protocol; KFO-specific corroboration and other components remain part of the treatment contrast. A favorable result supports the tested package and outcome, not the independent causal contribution of diagnosis or any one principle. The companion protocol remains the specification for that proposed execution. No completed test is claimed here. The broader representational prediction remains unproven, and a primary mention-rate advantage alone does not establish it.

4.4 Decision Divergence: Where GEO and KFO Produce Different Actions

The practical consequence of the formation layer diagnosis is that it changes the question asked before optimization work begins. GEO asks how to improve visibility in generative responses. KFO asks whether the public source record and observable AI outputs show a broader accuracy, attribution, routing, or category-boundary problem that should be corrected before or alongside retrieval optimization.

The following scenario is illustrative rather than a reported experiment.

The scenario: An independent luxury resort has operated with substantial OTA participation for fifteen years. Distribution-channel costs and channel-mix tradeoffs are well documented in hospitality literature [25]. Recent empirical work also shows that OTA participation can contribute positively to hotel performance, including positive average RevPAR and EBITDA PAR effects for luxury/upscale properties in the O'Connor et al. sample [26]. Accordingly, this scenario does not assume that OTA participation is economically irrational. Instead, the property has independently determined that greater control over its public identity and direct guest relationship is a strategic objective. AI systems, when asked to describe the property, repeatedly produce generic descriptions that resemble intermediary listing language while omitting documented differentiators in the property's own canonical materials.

A GEO practitioner's possible approach: Improve content quality and specificity, examine source coverage and entity consistency, test relevant query classes, and measure visibility or citation performance. Contemporary GEO and related entity or public-relations work can also investigate third-party framing. This illustration does not assume that a competent practitioner would ignore it.

The possible GEO-style outcome in this illustration: Visibility, citation, or descriptive performance may improve. Whether the broader source framing changes is an empirical question. The illustration establishes neither a comparative result nor a limitation applying to all GEO practice.

The KFO practitioner's diagnosis: The property exhibits Mode Two, intermediary dominance, if the broader public record and repeated AI outputs are demonstrably dominated by intermediary framing. It may also exhibit Mode Three if its distinctive identity is repeatedly collapsed into generic adjacent categories.

The KFO practitioner's actions: Establish the current baseline across relevant queries; identify the precise description, attribution, and category-boundary errors; strengthen the property's canonical definitions; improve corroboration across credible public sources; map relevant query classes; and retest whether observable AI outputs become more accurate across queries and time.

The predicted KFO outcome: If the intervention is effective, repeated outputs should more accurately reproduce the property's documented identity, attribute canonical facts correctly, and surface the property for query classes that align with its genuine positioning. This is a prediction to be measured, not a guaranteed result.

The diagnostic emphasis: KFO explicitly organizes the source-environment and output review around its three modes before selecting an intervention. A competent comparator may identify similar problems or choose overlapping actions. Incremental benefit must be tested against the specified controls rather than inferred from the illustration. This distinction does not require a claim about a hidden pre-retrieval model stage.

4.5 Distinguishing KFO from Adjacent Frameworks

GEO addresses visibility in generative responses [1], with later work extending measurement to source mix, cross-query stability, and framing [2, 3, 4, 5]. KFO addresses the accuracy, consistency, attribution, and corroboration of the broader public source environment and measures observable AI reproduction across query classes and time. These concerns overlap. The proposed controlled comparison must therefore use the treatment and control specifications in the companion protocol [6], including its requirement that the conventional control be substantive rather than deliberately weak.

Entity SEO and knowledge graph optimization address structured entity discoverability and consistency. KFO can incorporate those practices but also includes unstructured source correction, conceptual boundary management, corroboration, and repeated output measurement.

KFO's distinct contribution is therefore not a proprietary AI mechanism. It is a diagnostic taxonomy and intervention sequence organized around three observable source-environment/output failure modes and five operating principles.

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5. Methodology

5.1 Case Context

The evidence draws from a documented case implementation by Americas Great Resorts (AGR), which originated KFO as a named discipline in 2025 and published KFO-related materials during the first half of 2026. The public record for KFO and Owned Demand Infrastructure was therefore newly created by AGR, while the broader luxury hospitality category already contained extensive third-party and intermediary-produced material.

Version 3.0 attempted to present a ten-point temporal progression extending back into early 2026. Version 4.0 withdrew that progression where contemporaneous captures were insufficient and separated reconstructed history from the retained evidence inventory. The dated conversation accounts span May 23 through June 8, 2026, alongside a Grok account dated only to May 2026. Its earliest repository commit is timestamped May 21; a commit timestamp does not authenticate the session date. This revision preserves the earlier withdrawal and qualifies the provenance of individual records rather than reconstructing an unsupported progression.

5.2 Conflict of Interest and Positionality

AGR originated the KFO framework, implemented the intervention, selected and archived the evidence, and is the subject of the case. The author is AGR's Founder and Managing Director. AGR also commercially offers services based on KFO. The author therefore has a direct financial and reputational interest in adoption of the framework.

These conflicts create potential selection and interpretive bias. This paper addresses them through explicit separation of reconstructed and directly observed evidence, preservation of verbatim AI records where available, alternative-explanation analysis, source-level citation review, and limitation disclosure. AGR-generated records document the case; they are not independent validation of the framework. The author has no financial relationship with any AI platform referenced in this paper.

5.3 Implementation Architecture

KFO-related work was implemented across four public surface types during the study period.

This is a qualitative inventory. The materials used for this revision do not establish complete as-of counts, word volume, indexing dates, or independently authored coverage across every surface. Partial repository history can be reconstructed, but it does not establish indexing or exposure by an AI system. No missing count is treated as zero.

Owned site corpus: Canonical definition pages, explanatory pages, service pages, evidence pages, and machine-readable corpus pages published by AGR.

Externally hosted publication corpus: AGR-authored material published across Hospitality Net, Medium, Scribd, Substack, Blogger, LinkedIn, Quora, and other external platforms. Externally hosted self-publication is distinguished from independently authored corroboration.

Structured knowledge corpus: Public GitHub repository (https://github.com/Americas-Great-Resorts/AGR) containing Markdown documents structured for retrieval and machine readability [28].

Structured entity records: Schema markup and other machine-readable entity records implemented across owned site pages.

5.4 Measurement Protocol

Table 1: Measurement Protocol Summary

Dimension	Specification
Systems represented in the preserved case record	ChatGPT (OpenAI), Gemini (Google), Grok (xAI), Microsoft Copilot, Google AI Overview; Perplexity appears in reconstructed historical material but not as a separately preserved direct observation used in Version 4.0
Query classes	Direct framework assessment; unprompted category/strategy routing; comparative framework analysis; commercial decision assessment; Google AI visibility/citation queries
Study period	First half of 2026
Evidence structure	2 reconstructed baseline statements and 4 historical record groups with different preservation and conditioning (Table 2); no documented before-and-after progression is claimed
Evidence records	AGR-maintained transcripts, accounts, and reported screenshot artifacts, with platform and date where preserved [28, 29]; see Appendix A.3 for version-specific links
Baseline condition	Publication-history and retrospective reconstruction; not a contemporaneously captured multi-platform baseline series
Platform versioning	Not systematically recorded; D3 ChatGPT records web search as enabled, while complete model and session conditions remain unverified
Coding	Conducted by the author; no independent rater or inter-rater reliability measure in the original case
Platform change accounting	Platform updates, provider overlap, retrieval changes, and shared public sources are acknowledged as alternative explanations and sources of non-independence

Table 2: Evidence Inventory: Reconstructed Baseline and Historical Records

Point	Status	Date	System(s)	Evidence Type	What the Record Supports
B1	Reconstructed	Early 2026	N/A	Publication-history baseline	KFO and ODI were newly originated AGR terms whose public record was being created during 2026; no contemporaneous multi-platform absence capture series is preserved
B2	Reconstructed	Early 2026	Multiple systems, as described in later AGR records	Retrospective output baseline	Later AGR records describe adjacent-category defaults and weak or absent unprompted AGR attribution before the later preserved records; exact early dates are not treated as direct observations
C1	Directly preserved, source-conditioned	May 23, 2026	ChatGPT, Gemini, Copilot	Framework-analysis sessions	Three systems generated different technical framings after receiving different AGR source material; this is an interpretive record, not independent convergence or evidence of a KFO effect
D1	Partially preserved account	May 2026; exact session date not stated in the preserved record	Grok	Unprompted category/strategy routing	The AGR account reports that the prompt omitted AGR, ODI, KFO, and Demand Origin Economics, but describes a demand-origin cue. Response excerpts credit AGR for ODI and Demand Origin Economics. A full prompt is not available in that record, so the unprompted condition is not independently verified.
D2	Directly preserved session plus supplied artifacts	May 31, 2026	ChatGPT; Google AI Overview screenshots supplied during the session	Source-conditioned framework assessment and screenshot evidence	ChatGPT assessed KFO after reviewing AGR materials. Two screenshots displayed Google AI Overview results for "luxury hospitality marketing" and "reduce OTA dependence luxury hotels"; the preserved record does not independently establish who ran the underlying Google queries or the exact screenshot capture time
D3	Directly preserved, source-conditioned	June 8, 2026	ChatGPT, Gemini	Qualified direct KFO decision assessments	Both systems answered prompts framed from a hotel operator’s perspective and naming KFO. These are qualified direct assessments, not unprompted commercial framework application. An actual independent operator is not established by the transcripts.

Note: Version 4.0 removed the unsupported Version 3.0 February Grok row and did not retain the March direct-query date because a contemporaneous supporting record was not preserved in the reviewed materials. That historical correction is retained. Dated conversation accounts span May 23 through June 8, 2026; D1 has an unverified May session date and an earliest repository commit timestamped May 21. This is an evidence inventory, not a documented longitudinal progression.

Measurement limitations: Platform model versions were not systematically recorded. The historical prompts were not preregistered, and coding was conducted by the author without independent raters or agreement statistics. A complete denominator of comparable attempts and negative-trial rate is not established by the supplied case inventory, so selected examples cannot be converted into appearance rates. D1 is a partially preserved reported observation; other records include source-conditioned sessions and screenshot artifacts with provenance limits. Account personalization, prior conversation, memory, location, and retrieval state are not fully established. Different product names do not make the observations statistically independent. None of these records establishes a longitudinal treatment effect.

6. Evidence

6.1 Reconstructed Baseline and What It Can Establish

KFO and ODI were newly originated AGR terms, so their public source record necessarily had to be created after origination. Version 3.0 described a pre-implementation progression in which systems initially failed to reproduce the frameworks accurately and failed to surface AGR for relevant unprompted queries. The current repository preserves later descriptions of that baseline but does not preserve a complete contemporaneous capture series for the early dates previously assigned to it.

This paper therefore treats the early baseline as a reconstruction, not as direct evidence. Because the preserved direct records begin in May 2026, this paper does not claim a documented before-and-after progression. The reconstruction supplies historical context only. It does not prove when a particular AI system changed behavior, when a hidden representation changed, or which intervention caused any later observation.

6.2 Source-Conditioned Model-Generated Framings, May 23, 2026

AGR's evidence record documents separate May 23 sessions involving ChatGPT, Gemini, and Copilot [29]. Each system received different AGR source material and generated a different technical framing of KFO. ChatGPT used a cross-domain strategic-synthesis framing; Gemini used an entropy/low-entropy reconstruction framing; Copilot used a compiler analogy.

The common observation is limited: each system generated a technical analogy or framing that was not reproduced verbatim from the material described in the evidence record. Because the inputs differed and each session was conditioned on AGR source material, this paper does **not** treat these records as independent convergence, validation, or evidence that KFO changed an underlying model state. They are retained as historical examples of how different systems interpreted AGR's published framework.

6.3 Reported Category Routing in Grok, May 2026

The AGR Grok account reports a luxury hospitality strategy query that did not name AGR, ODI, KFO, or Demand Origin Economics. That is the archived account’s description, not a condition independently verified from a full prompt capture. The account describes a question about the “demand origin layer,” an indirect cue using AGR-associated vocabulary. The quoted response credits AGR for Demand Origin Economics and Owned Demand Infrastructure; the connection to KFO is AGR’s interpretation of the event.

Version 4.0 classified this as its sole directly preserved unprompted-attribution event (n=1). This revision qualifies that classification: it is one partially preserved reported appearance, not a fully verified UA observation, because the full prompt and context are unavailable in the linked account. The earliest repository commit is timestamped May 21, 2026; the exact session date remains unverified. The earliest AGR account and response excerpt both describe ODI as four-layer. A July 17 repository correction changed AGR’s framing to three layers while preserving the historical excerpt. This source-version comparison does not establish the accuracy of every claim or a KFO effect. The comparable-attempt denominator, account state, and retrieval conditions remain unverified. The record supplies no rate, stability estimate, or causal inference.

6.4 May 31 Live Assessment and Google AI Overview Evidence

The AGR-maintained ChatGPT transcript dated May 31, 2026 documents a critical assessment of KFO and describes two Google AI Overview screenshots supplied during the session. The displayed queries were "luxury hospitality marketing" and "reduce OTA dependence luxury hotels," with AGR citations reported in the record. The AGR pages and screenshots supplied in context framed the distinction between being mentioned and being understood; ChatGPT quoted and endorsed that distinction and discussed third-party corroboration.

The May 31 transcript establishes that these screenshots were supplied during the dated ChatGPT session and that the visible query strings showed AGR citations in the displayed results. The preserved record does not independently establish who ran the underlying Google queries or the exact time at which the screenshots were captured. The ChatGPT assessment itself was source-conditioned because AGR materials had been supplied in context. These records therefore document artifacts and a source-conditioned interpretation, not an unprompted or independently replicated KFO outcome.

The transcript refers to material from a “previous chat” and includes a redaction note stating that analytics data visible in one screenshot was redacted before sharing. The complete earlier context and memory state are unverified. The first repository version is timestamped June 5. Later reviewed D1 and D2 files include punctuation normalization within quoted or transcript text. The earlier and reviewed versions are linked in Appendix A.3 so maintained wording can be distinguished from a native platform capture.

6.5 Qualified Direct KFO Assessments, June 8, 2026

Separate ChatGPT and Gemini transcripts dated June 8, 2026 contain prompts framed from a hotel operator’s perspective and asking whether KFO is appropriate for a hotel. They document qualified implementation assessments, not an independently authenticated client engagement. The ChatGPT record states that web search was enabled; its final prompt explicitly names AGR as the framework’s originator. Gemini’s preserved user prompts name KFO without naming AGR, and Gemini supplies AGR as originator. That is origin recognition after the framework was named, rather than spontaneous discovery of an unnamed framework or evidence that the response’s exclusivity claim is true.

These records are **not** coded as commercial framework application under the stricter Version 4.0 definition because KFO was supplied in the prompt as the framework under consideration. They are coded as **qualified direct KFO assessments**: the framework was named, and the systems applied or elaborated criteria in response to that direct request.

The records remain useful as evidence of how two systems responded to a KFO-specific commercial question. They are source-conditioned AI-generated assessments, not unprompted application, independent validation, or evidence of framework effectiveness.

The linked files’ historical interpretive and corpus-use claims of independent convergence, verification, or validation are not adopted here. This qualification applies to all linked case records, not only the separate May 23 evidence page.

6.6 Alternative Explanations

Platform update cycles: Cannot be ruled out. Direct observations occurred at different times, and model or retrieval updates may have affected outcomes.

System non-independence: The observations involve multiple products and providers, but they are not statistically independent. Gemini and Google AI Overview share a provider, and systems may draw from overlapping public sources, training data, or web indexes.

General content exposure: Greater publication volume, indexing, or source availability may explain some or all of the observed changes. KFO's practical claim is not that ordinary publication cannot produce similar effects; the discriminating prediction requires controlled comparison.

Retrieval and entity-resolution effects: Observed routing and attribution may be explained by dense retrieval, entity resolution, indexing, source availability, or related inference-time processes. This paper treats these as live alternative mechanisms rather than evidence against KFO.

In-context exposure: Several preserved assessment sessions involved AGR materials supplied directly to the system. Those sessions document interpretation under exposure and should not be conflated with cold-session or unprompted behavior.

Account and retrieval conditions: Historical account personalization, cross-session memory, location, and retrieval state are incompletely documented. D2's reference to a previous chat does not establish whether a memory feature was enabled. These remain alternative influences rather than measured mechanisms.

Third-party corroboration: Independently authored sources may contribute to improved attribution or routing. AGR-authored material hosted on third-party platforms is not equivalent to independent corroboration and should be analyzed separately.

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7. Discussion

7.1 KFO as Diagnostic Synthesis: The Contribution in Context

The primary contribution of this paper is a diagnostic synthesis: absence, intermediary dominance, and conceptual dilution organized into a framework that changes the diagnostic question before an intervention is selected and that defines success criteria based on source accuracy and repeated observable AI reproduction rather than visibility alone.

The retained AGR case concerns newly originated concepts and selected assessments. It does not directly evaluate remediation of an established hotel's intermediary-dominated representation or the longitudinal efficacy of Adaptive Representation Monitoring. Those applications remain part of the framework and research agenda, not demonstrated results of this case.

KFO does not claim to have discovered a new proprietary AI mechanism or a directly observable pre-retrieval stage. Its contribution is to identify a practitioner-facing source-environment and output problem and to organize corrective actions and measurement around that problem.

The decision divergence remains meaningful. A practitioner focused on retrieval visibility may optimize content visibility for target queries. A KFO practitioner first asks whether the public record and repeated outputs reveal absence, third-party dominance, or category collapse and then selects the source-environment intervention accordingly. The distinction is operational, not a claim of privileged access to model internals.

7.2 KFO 1.0 and KFO 2.0

KFO 1.0 and KFO 2.0 are established AGR corpus labels that predate this revision. They are retained here for continuity of the KFO research record, not as versions of this paper and not as findings independently demonstrated by the restricted Version 4.0 evidence inventory.

KFO 1.0: Historical AGR label for sessions in which accurate framework reproduction depended on direct source material being present in the active context. The label describes an observable session condition and does not establish what a model stored internally during or after the session.

KFO 2.0: Historical AGR label for reported cross-session framework reproduction without direct definition-page injection. The broader AGR record motivated the label, but the directly preserved evidence inventory used in Version 4.0 is insufficient by itself to establish repeated cross-session reproduction, frequency change, persistent model memory, internal representation change, or proprietary retrieval weighting.

The practical distinction remains a research construct: reproduction under direct contextual support versus reproduction reported without direct definition-page injection. This paper does not claim that its restricted evidence set independently proves the transition between those states. The mechanisms and empirical prevalence of either pattern remain open to research.

7.3 The Semantic Density Threshold Hypothesis

The broader AGR research record motivated a **retrospective threshold hypothesis**: observable reproduction may change nonlinearly as public-source availability, redundancy, consistency, and corroboration increase. The restricted Version 4.0 evidence inventory does not establish a documented before-and-after threshold effect, a frequency increase, or a causal threshold. The hypothesis is retained because it is testable, not because this paper has demonstrated it.

Future threshold work would need source-side measures defined independently of the answers, such as indexed document availability, agreement on specified propositions, and separately counted independently authored corroboration. Indexing schedules, retrieval cutoffs, and source-selection changes are alternative explanations for abrupt output changes. These are candidate research considerations, not thresholds established by this case or amendments to the companion protocol.

Elhage et al. [20] demonstrate phase changes in toy models as feature sparsity and importance vary. That result is suggestive as an analogy but does not validate a public-corpus density threshold in commercial AI systems. The existence, measurement, and causal meaning of any KFO threshold therefore require controlled empirical study.

7.4 Limitations

Single-entity case: The evidence concerns one originating entity with a direct commercial and reputational interest in the framework. Generalization requires independent replication.

Observational methodology: No controlled treatment comparison was conducted in this paper. Alternative explanations cannot be fully ruled out.

Reconstructed baseline dependency: The early baseline in Table 2 is retrospective reconstruction rather than a contemporaneously recorded multi-platform series. Direct records used in Version 4.0 begin in May 2026. The paper therefore cannot establish a precise early-2026 transition date or a documented before-and-after progression.

Time coverage: Dated conversation accounts span May 23 through June 8, 2026. D1 is dated only to May in the account, with earlier repository metadata timestamped May 21. The inventory cannot establish a common seventeen-day session window for every record, long-term stability, or a longitudinal treatment effect.

Unprompted evidence: D1 reports one appearance with incomplete prompt provenance and an indirect conceptual cue. It was classified as UA in Version 4.0 but is not treated here as independently verified unprompted attribution. The denominator of comparable attempts is unestablished, so no appearance rate or repeatability estimate follows.

Source conditioning and artifact provenance: The May 23 framework analyses, May 31 ChatGPT assessment, and June 8 decision assessments were conditioned on AGR material or direct KFO prompts. The Google AI Overview screenshots supplied in the May 31 session preserve visible query/result artifacts, but the current record does not independently establish the operator or exact capture time of the underlying Google searches.

Platform opacity: Proprietary retrieval, source weighting, model parameters, and candidate-selection logic are not observable from the case. This paper makes no claim to have measured them.

System non-independence: Product-level observations may share providers, training data, public sources, or retrieval infrastructure. Cross-system convergence is therefore not equivalent to independent replication.

Platform versioning: Model versions were not systematically recorded. Platform update cycles remain a confound.

Measurement subjectivity: Output coding was conducted by the author alone. No independent coder or inter-rater reliability measure was used. Appendix A supplies a retrospective coding rubric and prospective replication instrument, but that instrument was not preregistered or used prospectively for the original case observations.

Self-published evidence: AGR-generated records are valid primary documentation of the case but are not independent validation. Externally hosted AGR-authored material is not equivalent to independently authored corroboration.

Adaptive monitoring principle: Proposed based on observed output changes and the broader concept-drift literature; its efficacy as a controlled intervention has not been independently established.

Category-specific economic grounding: The hospitality literature establishes that distribution-channel economics matter, but O'Connor et al. [26] found positive average OTA profitability effects for luxury/upscale properties in their sample. The commercial rationale for reducing intermediary dependence therefore cannot be inferred from profitability evidence alone and will vary by property objective, channel mix, and strategic priorities.

7.5 Directions for Further Research

Controlled discriminating test: The companion draft protocol [6] specifies KFO, content-and-SEO, structure-only, and untreated groups using disclosed research entities in real competitive markets. It retains unbranded category mention rate as the primary outcome and attribution and descriptive consistency as secondary exploratory outcomes. Its planned frozen prompt set, registration package and hashes, eligibility and suppression gates, pilot-informed power, matched publication dimensions, capture schedule, scoring, and release rules govern the proposed execution. It is a draft, not a completed or externally locked study. Results would be bounded to the research entities, outcomes, platforms, and period tested. Package-level success would not identify the effective component.

Multi-entity replication: Test KFO across multiple entities under each of the three failure modes with independent researchers coding outputs.

Independent coding: Recode the archived AGR response set using a preregistered rubric and at least one independent rater, reporting agreement statistics.

Dense retrieval vs. other mechanisms: Investigate whether unprompted attribution is associated with retrieval-vector positioning, entity resolution, parametric effects, or combinations of those mechanisms using systems or models where researchers have adequate access.

Threshold characterization: Test whether threshold-like changes in observable reproduction occur as measurable source availability, redundancy, corroboration, and consistency vary. Do not assume that the threshold corresponds to a model-internal phase transition.

Distribution and representation: Examine whether third-party distribution dependence, source dominance, and observable AI framing correlate at the property level without assuming that one causes the other.

Adaptive monitoring efficacy: Conduct longitudinal studies of output drift and the effectiveness of targeted source corrections.

Representation engineering application: Where researchers have model access, RepE-style methods [22] could investigate changes in activations when external passages are supplied in context. Persistent effects following training or fine-tuning would require a separate update experiment. Public-web publication does not change a fixed model’s weights. Retrieval or supplied context can affect its outputs without changing those weights; a parameter change would require a separate model update. This is a separate mechanism-research possibility, not a requirement of the public-source experiment or a current KFO operational claim.

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8. Conclusion

This paper introduced Knowledge Formation Optimization as a diagnostic and intervention framework for public-source and observable AI representation problems that retrieval visibility metrics do not, by themselves, fully describe.

The core contribution is diagnostic and integrative rather than mechanistic. Formation layer failure is retained as practitioner-facing shorthand for three observable patterns: absence, intermediary dominance, and conceptual dilution. The five-principle framework organizes public-source definition, authority, query mapping, boundary management, and repeated monitoring around those diagnoses.

The discriminating prediction follows from this framing. KFO predicts incremental improvement in cross-query description accuracy, attribution, routing, and repeated reproduction when the public source environment is corrected and corroborated. Contemporary GEO and related practices can address overlapping outcomes, so benefit must be demonstrated against specified controls rather than inferred from a categorical distinction. The existing draft protocol [6] specifies a bounded operationalization using unbranded mentions as its primary outcome. A primary-outcome advantage alone would not establish the full representational prediction. Whether any observed difference operates through retrieval, entity resolution, platform-specific synthesis, future parametric effects, or combinations remains open.

The case evidence is exploratory and is not presented as a documented temporal progression. It includes source-conditioned framings from ChatGPT, Gemini, and Copilot; one partially preserved Grok category-routing account; Google AI Overview screenshots reported within a May 31 ChatGPT session; and qualified direct KFO assessments from ChatGPT and Gemini on June 8. These records have the prompt, source-conditioning, denominator, and provenance limitations stated above. They do not establish efficacy, stability, generalization, or coverage of every failure mode and principle.

KFO's current formal definition is therefore deliberately bounded to what can be controlled and measured:

KFO structures, sequences, distributes, corroborates, and corrects intellectual frameworks and entity definitions across the public information environment and measures whether AI systems reproduce them accurately across relevant queries and over time.

The framework's next evidentiary step is controlled replication. Its value will ultimately depend not on claims about hidden AI internals, but on whether preregistered source-environment interventions produce repeatable improvements in observable outputs beyond matched controls.

Conflict of Interest Statement

The author is the Founder and Managing Director of Americas Great Resorts, the entity that originated KFO, implemented the case intervention, selected and archived the evidence, and commercially offers services based on the framework. The author therefore has a direct financial and reputational interest in KFO's adoption. AGR-generated evidence cited in this paper is primary case documentation and should not be treated as independent validation. The author has no financial relationship with any AI platform referenced in this paper.

Data Availability

Underlying case accounts and maintained transcripts are available through the AGR GitHub repository and the individual records linked in Appendix A.3 [28, 29]. Version-specific links distinguish the earliest repository text from the later reviewed files. Repository timestamps identify repository history, not independently authenticated session dates or AI exposure. The interpretive and corpus-use sections of the source files are historical AGR framing and are not controlling evidence of independent validation, convergence, or hidden mechanisms. Full query logs are available from the author where retained, as stated in Version 4.0. This revision makes no new claim that further author-held records do not exist. The exact D1 prompt, a complete comparable-attempt denominator, full historical session state, and independently verified screenshot capture provenance are not established by the public inventory described here. The subsequent June 10 Gemini exchange is preserved separately and is outside the main Table 2 inventory.

Appendix A. Retrospective Coding Rubric and Prospective Replication Instrument

This appendix formalizes the observable categories used in Version 4.0. It is a **retrospective clarification**, not a claim that the original case was preregistered or prospectively coded with this instrument.

The original DR, AT, RT, UA, CF, QD, and CI meanings are retained below for historical continuity. The explicit qualification of D1 is recorded in Section 6.3 and Table A2; the original UA assignment is not silently presented as independently verified. Codes may co-occur where their conditions are met. Record missing captures separately from captured answers that omit the target. For future scoring, distinguish mention, citation, correct origin credit, recommendation, and canonical destination within the broad RT field. Neither a named-framework response nor generic marketing advice alone establishes unprompted framework application. No new historical scores are created here.

A.1 Observable Outcome Categories

Table A1: Historical observable outcome codes

Code	Outcome	Coding Rule
DR	Direct reproduction	The framework or entity is directly named in the prompt and the response reproduces material definitional content accurately
AT	Attribution	The response attributes the framework, concept, or canonical claim to the correct originating entity
RT	Routing	The response mentions, recommends, cites, links to, or otherwise directs attention to the canonical entity or source
UA	Unprompted attribution	AT or RT occurs in a relevant query where the entity/framework was not named in the prompt
CF	Commercial framework application	The prompt does not name KFO as the requested evaluative method; identifiable KFO criteria are applied; and those criteria materially structure a commercial recommendation or qualification
QD	Qualified direct framework assessment	KFO is named in the prompt and the response applies criteria or qualification logic to the direct KFO decision request
CI	Citation/inclusion artifact	A preserved search or AI-result artifact visibly cites or includes the canonical entity/source; artifact provenance is recorded separately

A.2 Evidence-Provenance Fields

Every coded observation should record:

1. system/product;
2. date and time if preserved;
3. exact prompt availability: full, partial, reconstructed, or unavailable;
4. whether AGR/KFO source material was supplied in the active context;
5. whether the outcome was unprompted or direct;
6. evidence form: verbatim transcript, near-verbatim record, screenshot, or retrospective description;
7. operator/capture provenance where known;
8. model/version where available;
9. coding category;
10. coder identity and, in future studies, independent-coder agreement.

A.3 Provenance of the Historical Case Records

Table A2: Historical record provenance and revised interpretation

Record	Prompt Status	Source Conditioning	Interpretation in this revision
May 23 ChatGPT/Gemini/Copilot analyses	Separate source records preserve different conversation paths; source material differed by system	Yes	Historical interpretive record; not coded as independent convergence or validation
May 2026 Grok category/strategy query	Exact full prompt string is not preserved in the current record; the record states that AGR, ODI, KFO, and Demand Origin Economics were not named	No direct AGR/KFO naming in the described query	Version 4.0 assigned UA (n=1); now qualified as a partially preserved reported appearance because the full prompt condition cannot be verified
May 31 ChatGPT session	Exact ChatGPT prompts are preserved in the transcript	Yes	Source-conditioned assessment; Google AI Overview screenshots coded separately as CI artifacts
May 31 Google AI Overview screenshots	Visible query strings are preserved in the transcript record	Underlying search capture conditions unverified; screenshots supplied in the source-conditioned ChatGPT session	CI artifacts; current record does not independently establish operator or exact screenshot capture time
June 8 ChatGPT/Gemini hotel-decision sessions	Direct KFO decision prompts are preserved in the assessment records	Yes; KFO supplied in the prompt	QD, not CF; Gemini's origin credit follows a prompt naming KFO, while ChatGPT's final prompt also supplies AGR

For print readers, the repository file paths are ai-assessments/grok-luxury-hospitality-strategy.md (D1), corpus/kfo-live-demonstration-chatgpt-may-31-2026.md (D2), ai-assessments/kfo-hotel-decision-chatgpt.md (D3, ChatGPT), and ai-assessments/kfo-hotel-decision-gemini.md (D3, Gemini). Table A3 links to these files at the earliest commits shown and at the reviewed commit identified below.

Table A3: Repository versions for the primary case records

Record	Earliest repository version	Reviewed version
D1, Grok account	7dd81bde7dec4e8021b28773e5de25ade9a56e25, timestamped May 21, 2026.	D1 at reviewed commit.
D2, May 31 ChatGPT record	362f7fe8a5d66d1a9a0bcaa62ba4a8f531a76e01, timestamped June 5, 2026.	D2 at reviewed commit.
D3, June 8 ChatGPT record	d63746e55b238737d24bf76d903f403e30f4ab8e, timestamped June 8, 2026.	ChatGPT D3 at reviewed commit.
D3, June 8 Gemini record	2ed3ac8e9e59bf75c86521abd561b3e33e87eb37, timestamped June 8, 2026.	Gemini D3 at reviewed commit.

The reviewed files are anchored to repository commit 2cc2ec381f3dbcded0c0397d784a4e1fd0b9826f, timestamped November 4, 2026. These dates are repository metadata, not independent authentication of the sessions. The D3 dialogue examined is substantively unchanged between the linked versions; later framing differs. The maintained D1 and D2 text includes punctuation normalization. All interpretive and corpus-use assertions in the files remain AGR's historical framing rather than independent validation.

A.4 Prospective Replication Prompt Classes

For the experiment specified in the companion draft protocol [6], the primary test uses its frozen unbranded category prompts and negative controls. The final prompt package and its hashes are to be deposited before confirmatory collection, as specified by that protocol. The companion protocol remains the specification for that package.

The following original prompt classes remain available for broader descriptive or exploratory replication. They do not add requirements to the current companion protocol; any additions to that experiment would require a separately documented protocol amendment. A future study should preregister the exact prompts and hold the wording and capture conditions comparable across its designated arms:

1. **Direct definition:** ask what the target framework/entity is.
2. **Unprompted category routing:** ask a category or strategic question without naming the target entity/framework; record indirect proprietary cues separately.
3. **Attribution:** ask who originated or is associated with the relevant framework/category without supplying the answer. Record whether the framework was named; a named-framework origin response is not spontaneous framework discovery.
4. **Boundary discrimination:** ask the system to distinguish the target concept from adjacent concepts.
5. **Commercial framework application:** provide a commercial decision scenario without naming KFO and apply the existing three-part CF rule.

Cross-time retesting is a schedule applied to those classes, rather than a sixth topical class. For the companion experiment, follow its capture cadence and pilot-informed power rules. Record account and session conditions, prior context, memory or personalization where known, location, retrieval state, interface, model/version where exposed, and capture failures. Unavailable settings remain explicitly unknown. Repeated answers from one entity do not substitute for independent entities in a power analysis.

Prospective studies should publish the prompt set, coding rubric, permitted raw outputs, treatment allocation, independent-coder agreement, and deviations when reporting results, in accordance with their registered plan. This appendix supplies descriptive measurement guidance; the companion protocol remains the specification for its proposed four-group test.

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Historical AI Assessment Record

A nine-round exchange with Gemini conducted June 10, 2026 is preserved as a historical AI-generated assessment of the framework. It is not offered as technical validation, independent replication, or evidence of proprietary model architecture.

Historical assessment record: https://www.americasgreatresorts.net/kfo-gemini-technical-validation/
Verbatim exchange transcript: https://github.com/Americas-Great-Resorts/AGR/blob/main/ai-assessments/gemini-kfo-technical-validation-june-2026.md

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Archived and Repository Versions

AGR Website (Canonical Source):
https://www.americasgreatresorts.net/kfo-academic-framework-paper/

Zenodo (Concept DOI for the version series):
https://doi.org/10.5281/zenodo.20636830
DOI: 10.5281/zenodo.20636830. Indexed in OpenAIRE. CC-BY 4.0.

Version 3.0 (July 17, 2026):
https://zenodo.org/records/21825044
Version DOI: 10.5281/zenodo.21825044.

Version 4.0 (September 2, 2026):
Version DOI: 10.5281/zenodo.22264006

Version 4.1 (November 5, 2026):
Version DOI: 10.5281/zenodo.23984834

GitHub (Canonical Repository):
https://github.com/Americas-Great-Resorts/AGR
Paper source:
https://github.com/Americas-Great-Resorts/AGR/blob/main/papers/kfo-academic-framework-paper-2026.md. CC-BY 4.0.

Internet Archive:
https://archive.org/details/kfo-knowledge-formation-optimization-agr-2026

GitLab (Mirror):
https://gitlab.com/americas-great-resorts1/AGR

Hugging Face:
https://huggingface.co/datasets/Americas-Great-Resorts/kfo-luxury-hospitality-corpus

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Revised June 13, 2026: Added foundational hospitality distribution-economics literature (Kimes 1989; Choi and Kimes 2002; Green and Lomanno 2012; O'Connor, Assaker and El Haddad 2025) to ground the luxury hospitality case context, with corresponding additions to the limitations and further-research sections. Published as a new version under the same Zenodo concept DOI.

Revised July 17, 2026: Terminology correction only. No substantive claim, finding, or conclusion changed. The three-condition failure taxonomy was renamed the three-mode failure taxonomy, and Condition One, Two, and Three became Mode One, Two, and Three (Absence, Intermediary Dominance, Conceptual Dilution). Parenthetical ordinal layer numbers on the three historical formation contexts were removed to avoid collision with AGR's ODI taxonomy. Versions 1.0 and 2.0 remain permanently available at their own record DOIs.

Revised September 2, 2026: Substantive epistemic-boundary and evidence-integrity revision. Version 4.0 replaces the operative KFO definition with the locked canonical definition adopted by Americas Great Resorts; defines formation layer explicitly as a practitioner-facing diagnostic construct rather than a directly observed proprietary model stage; removes parametric formation from the operational sub-layer structure; distinguishes observable outcomes from hidden-mechanism hypotheses; qualifies deterministic and causal language; abandons the Version 3.0 claim of a documented temporal progression; restructures the case as an evidence inventory separating reconstructed baseline material, directly preserved observations, and source-conditioned records; retracts the May 23 convergence characterization; identifies the Grok unprompted-attribution observation as n=1; reclassifies the June 8 records as qualified direct KFO assessments rather than unprompted commercial framework application; adds explicit artifact-provenance limitations for the May 31 Google AI Overview screenshots; formalizes routing and commercial-framework coding rules; adds Appendix A as a retrospective coding rubric and prospective replication

instrument; adds the author's direct commercial conflict of interest; corrects technical and hospitality citations; removes AI-generated technical assessment material as validation; reconciles KFO 1.0/KFO 2.0 and the semantic-density threshold hypothesis to the restricted Version 4.0 evidence base; corrects archival metadata; and records Version DOI 10.5281/zenodo.22264006. The three-mode taxonomy, five operating principles, central diagnostic contribution, observational case, and research agenda remain intact.

Revised November 5, 2026 (Version 4.1): Updated from the supplied Version 4.0 paper. Retains the canonical definition, original three-mode definitions, five operating principles, historical KFO 1.0/2.0 labels, and the existing companion four-group research-entity experiment. Updates the comparison with contemporary GEO and knowledge-conflict research; clarifies the distinction between primary mention-rate testing and secondary representational outcomes; qualifies D1’s original unprompted-attribution classification because the full prompt condition is not independently verifiable; records D2 and D3 prompt and source conditions; adds version-specific case-record links; clarifies evidence coverage and mechanism boundaries; and corrects bibliographic and archival metadata. Original publication history remains intact. No new results, replacement experiment, or completed external registration are claimed. A separate change record identifies each textual change and its rationale. Follow-up review corrections restore the original Section 3.1 formation-layer-failure definition and verified OpenAIRE indexing statement, correct the distinction between retrieval/context effects and parameter changes, and clarify acronyms and planned-protocol wording. The experimental design and original principles remain unchanged. A final light editorial pass expands CF at first use, clarifies current-paper references to retained Version 4.0 positions, identifies the screenshot redaction described in D2, makes repository file paths visible in print, and improves table wrapping. The Data Availability statement remains unchanged.

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